Excel Data Analysis

Task 5 · Data Analytics Track · Level 1 – Day 5

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Total Sales	Total Profit	Profit Margin	Total Orders	Avg Order Value
406,637	17,520	4.3%	600	678

Note: Superstore-style sample dataset (600 orders, Jan–Dec 2025) used for this analysis.

1. Objective

Analyze a structured retail sales dataset end-to-end using core Excel tools – formulas (SUMIFS/COUNTIFS/AVERAGEIFS), sorting, filtering, pivot-style summaries and basic KPIs – to answer business questions about sales and profitability.

2. Dataset & Workbook Structure

Columns: Order ID, Order Date, Region, Category, Sub-Category, Segment, Quantity, Sales, Discount, Profit. The workbook has three tabs: **Raw_Data** (untouched source, frozen header, filters, named ranges), **Working** (all formulas and pivot-style tables) and **Summary** (KPIs, charts and key findings).

3. Approach

(1) Loaded data into Raw_Data and froze headers. (2) Created named ranges (Sales, Profit, Region, Category, Segment, OrderDate...). (3) Built KPIs and five summary tables using SUMIFS/COUNTIFS/AVERAGEIFS. (4) Added charts on the Summary tab. (5) Wrote findings and recommendations.

4. Business Questions Answered

Pivot 1 – Sales by Category

Category	Sales	Profit	Margin	Orders	Avg Order Val	% of Sales
Furniture	143,184	-15,860	-11.1%	146	981	35.2%
Office Supplies	42,325	2,422	5.7%	294	144	10.4%
Technology	221,127	30,958	14.0%	160	1,382	54.4%
Grand Total	**406,637**	**17,520**	**4.3%**	**600**	**678**	**100.0%**

Pivot 2 – Sales by Region

Region	Sales	Profit	Margin	Orders	Avg Order Val	% of Sales
Central	76,209	-355	-0.5%	119	640	18.7%
East	125,752	4,003	3.2%	172	731	30.9%
South	81,709	7,278	8.9%	113	723	20.1%
West	122,967	6,594	5.4%	196	627	30.2%
Grand Total	**406,637**	**17,520**	**4.3%**	**600**	**678**	**100.0%**

Pivot 3 – Sales by Customer Segment

Segment	Sales	Profit	Margin	Orders	Avg Order Val	% of Sales
Consumer	192,372	7,065	3.7%	291	661	47.3%
Corporate	114,809	3,556	3.1%	168	683	28.2%
Home Office	99,456	6,899	6.9%	141	705	24.5%
Grand Total	**406,637**	**17,520**	**4.3%**	**600**	**678**	**100.0%**

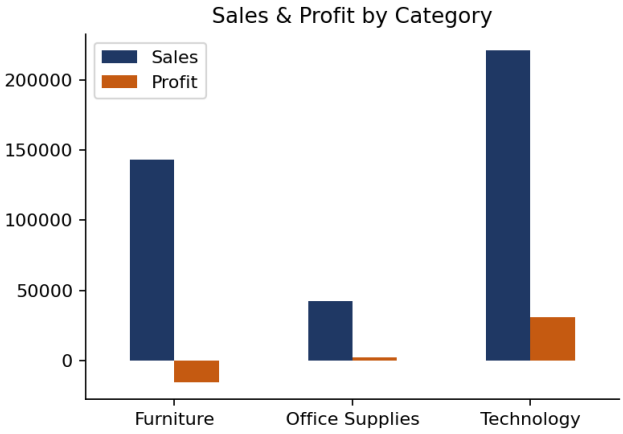
Pivot 4 – Monthly Sales Trend

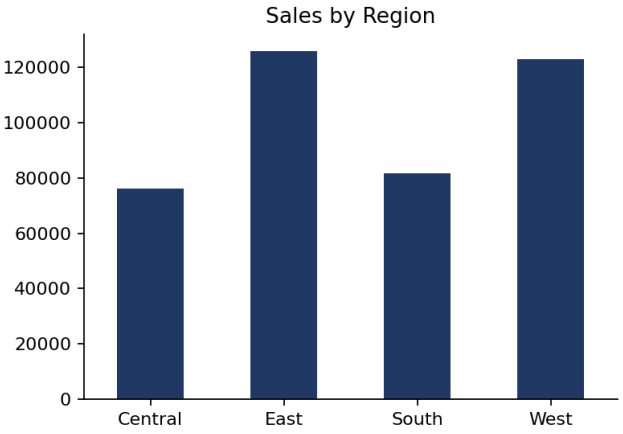
Month	Sales	Profit	Margin	Orders
Jan	24,634	2,081	8.4%	36
Feb	24,472	879	3.6%	38
Mar	47,504	2,962	6.2%	53
Apr	22,901	53	0.2%	32
May	40,829	2,487	6.1%	49
Jun	23,751	1,489	6.3%	34
Jul	31,080	3,008	9.7%	45
Aug	24,733	1,363	5.5%	43
Sep	34,010	1,107	3.3%	59
Oct	27,514	160	0.6%	54
Nov	54,991	1,006	1.8%	87
Dec	50,219	925	1.8%	70
Total	**406,637**	**17,520**	**4.3%**	**600**

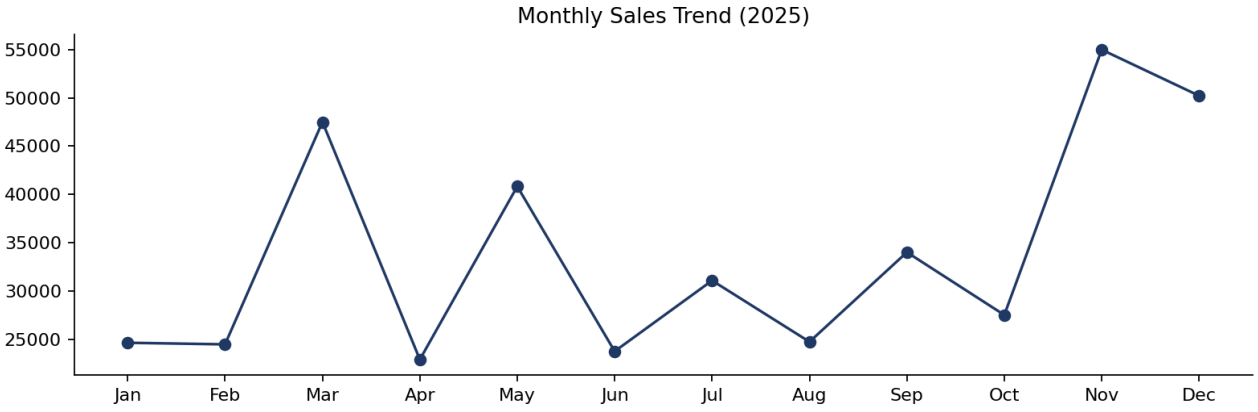
Pivot 5 – Sales: Category × Region

Category	Central	East	South	West	Total
Furniture	24,016	53,168	15,045	50,955	143,184
Office Supplies	8,531	13,508	6,777	13,509	42,325
Technology	43,662	59,076	59,886	58,502	221,127
Total	**76,209**	**125,752**	**81,709**	**122,967**	**406,637**

5. Charts







6. Key Findings

The business generated about 406.6K in sales and 17.5K in profit across 600 orders in 2025, a thin overall margin of 4.3%. Technology is the star category (54% of sales, 14.0% margin), while Furniture contributes 35% of sales but loses money (-11.1% margin, about -15.9K), mainly because of heavy discounting. Orders discounted 30% or more brought in 72K of sales but produced a combined loss of about 14K. East and West lead on sales, whereas the Central region is roughly break-even (-0.5% margin) and South has the best margin (8.9%). Sales peak in Nov-Dec (about 105K together) and March, so inventory and marketing should be planned around those periods.

7. Recommendations

- Cap discounts on Furniture and high-discount orders.
- Push Technology products, the most profitable category.
- Review pricing and costs in the Central region.
- Plan stock and campaigns for Nov–Dec peak season.

8. Interview Questions

SUMIFS vs PivotTable? Use SUMIFS for fixed, formula-driven report cells that feed dashboards and must update automatically; use a PivotTable for quick exploration, slicing and drag-and-drop analysis of the same question.

Keeping a PivotTable's source live? Convert the data to an Excel Table (Ctrl+T) or use a dynamic named range, then use Refresh All so new rows are included.

Retail KPI? Profit margin – it shows whether sales growth is actually profitable. Also useful: average order value and month-over-month sales growth.

9. Tools Used

Microsoft Excel (SUMIFS, COUNTIFS, AVERAGEIFS, named ranges, charts).